(a) Given the congruances are,

$$x \equiv 1 \pmod 3$$
$$x \equiv 2 \pmod 5$$
$$x \equiv 3 \pmod 7$$

Moduli, $M = 3 \times 5 \times 7 = 105$

For $m_1 = 3$, $M_1 = 105/3 = 35$

$\therefore M_i \times y_i \equiv 1 \pmod{m_i}$

$35 \times y_1 \equiv 1 \pmod 3$

$35 \bmod 3 = 2$

$\therefore 2 \times y_1 \equiv 1 \pmod 3$

$\therefore y_1 = 2$, as $2 \times 2 \equiv 1 \pmod 3$

for $m_2 = 5$; $M_2 = 105/5 = 21$

$21 \times y_2 \equiv 1 \pmod 5$

$1 \times y_2 \equiv 1 \pmod 5$

$\therefore y_2 = 1$

for $m_3 = 7$; $M_3 = 105/7 = 15$

$15 \times y_3 \equiv 1 \pmod 7$

$\Rightarrow 1 \times y_3 \equiv 1 \pmod 7$

$\therefore y_3 = 1$

$\sum M_i y_i = 1 \times 35 \times 2 + 21 \times 1 + 15 \times 1 = 70 + 42 + 45 = \cancel{105}\ 157$

$\therefore x \equiv 157 \pmod{105}$ | $x \equiv 52 \pmod{105}$

(b) Congruences, $x \equiv 5 \pmod{11}$, $x \equiv 14 \pmod{29}$, $x \equiv 15 \pmod{31}$

$\therefore$ Moduli $M = 11 \times 29 \times 31 = 9889$

$\therefore$ final Solution will be modulo 9889

$M_1 = \frac{9889}{11} = 899$, $M_2 = \frac{9889}{29} = 341$, $M_3 = \frac{9889}{31} = 319$

$\therefore$ Inverses are, $M_i Y_i = 1 \pmod{m_i}$

$\therefore$ For, $m_1 = 11$, $899 \times Y_1 \equiv 1 \pmod{11}$

$\Rightarrow 8 \times Y_1 \equiv 1 \pmod{11}$

$\Rightarrow \therefore Y_1 = 7$ as $8 \times 7 = 56$ and mod 11 is $= 1$

$11 \times 81 = 891$

$\therefore 899 - 891 = 8$

For $m_2 = 29$, $341 \times Y_2 \equiv 1 \pmod{29}$

$\Rightarrow 22 \times Y_2 \equiv 1 \bmod 29$

$\therefore Y_2 = 4$

For $m_3 = 31$ $319 \times Y_3 \equiv 1 \pmod{31}$

$\Rightarrow 9 \times Y_3 \equiv 1 \pmod{31}$

$\therefore Y_3 = 7$

$\therefore$ Sum $= 5 \times 899 \times 7 + 14 \times 341 \times 4 + 15 \times 319 \times 7 = 84056$

$\Sigma a_i M_i Y_i$

$a \rightarrow$ remainder

$\therefore x \equiv 84056 \pmod{9889}$

$\therefore x \equiv 4944 \pmod{9889}$

(c) Congruences; $x \equiv 5 \pmod 6$, $x \equiv 4 \pmod{11}$, $x \equiv 3 \pmod{17}$

$\therefore M = 6 \times 11 \times 17 = 1122$

$\therefore M_1 = \frac{1122}{6} = 187$, $M_2 = \frac{1122}{11} = 102$, $M_3 = \frac{1122}{17} = 66$

For, $m_1 = 6$ $\quad \therefore 187 \times y_1 \equiv 1 \pmod 6$

$\Rightarrow 1 \times y_1 \equiv 1 \mod 6$

$\therefore y_1 = 1$

$m_2 = 11$. $\quad 102 \times y_2 \equiv 1 \pmod{11}$

$\therefore 3 \times y_2 \equiv 1 \pmod{11}$

$\therefore y_2 = 4$

$m_3 = 17$ $\quad 66 \times y_3 \equiv 1 \pmod{17}$

$\Rightarrow 15 \times y_3 \equiv 1 \pmod{17}$

$\Rightarrow \therefore y_3 = 8$

$\therefore$ Sum $= 5 \times 187 \times 1 + 4 \times 102 \times 4 + 3 \times 66 \times 8 = 4151$

$\therefore x \equiv 4151 \pmod{1122}$

$\therefore x \equiv 785 \pmod{1122}$